

Dynamic Guideline: WHO-Based Snake Bite - Diagnosis and/or Treatment and/or Management

I. WHO Latest Guideline Summary (Summary Date: 29 April 2026)

1.1 Guideline Overview

Scope note

For snakebite envenoming, the latest WHO **clinical management guideline** identified is the WHO Regional Office for South-East Asia document **“Guidelines for the management of snakebites, 2nd edition,”** published on **16 August 2016** [WHO publication page](#). Since then, WHO has issued or reported additional snakebite-related strategic and technical guidance, especially on antivenom quality, target product profiles, and novel therapeutics, but these are not full bedside clinical management guidelines.

The requested title **“Meaningful youth engagement in a WHO guideline development process: case study of WHO [] guideline”** was identified as a WHO evidence-brief/webinar linked to the **WHO Abortion care guideline (2022)**, not snakebite envenoming [WHO webinar page](#). No snakebite-specific WHO youth-engagement guideline-development case study was identified in the supplied evidence.

Exactly 5 WHO snakebite-related guidelines / technical guidance documents synthesized

No.	WHO document / guidance	Date / status identified	Role in this dynamic guideline
1	Guidelines for the management of snakebites, 2nd edition	16 August 2016	Latest identified WHO clinical management guideline for diagnosis, first aid, antivenom use, monitoring, referral, and health-system organization WHO publication page .
2	WHO guidance on production, control and regulation of snake antivenom immunoglobulins	WHO technical guidance page identified; date not extractable from supplied snippets	Antivenom quality, production, regulation, and the role of antivenom as the only specific treatment for envenoming WHO antivenom immunoglobulins guidance page .
3	Snakebite Envenoming: A Strategy for Prevention and Control	2019 strategy cited in supplied evidence	Global roadmap aiming to reduce snakebite mortality and disability by 50% by 2030, structured around community engagement, safe treatment, health-system strengthening, and partnerships WHO fact sheet .
4	WHO target product profile guidance for snakebite	First WHO snakebite-treatment guidance	Product-development standards for safe, effective, context-appropriate

No.	WHO document / guidance	Date / status identified	Role in this dynamic guideline
	antivenoms	reported 23 June 2023 ; one snakebite antivenom TPP reported released in 2024	antivenoms WHO 2023 news item , Global report on neglected tropical diseases 2025 .
5	Target product profiles for small molecule or engineered antibody therapeutics for treatment of snakebite envenoming	31 December 2025 publication page; WHO news 27 February 2026	WHO's first public-benefit TPPs for novel snakebite therapeutics, including small molecules and engineered antibody biologics WHO publications page , WHO news .

Guideline Title

Guidelines for the management of snakebites, 2nd edition

Publication Date

16 August 2016

Target Population

- People bitten by snakes, especially in rural and agricultural communities.
- High-risk groups include:
 - Agricultural workers.
 - Children, who may have more severe effects because of smaller body mass.
 - Rural populations with delayed access to care.
- WHO estimates approximately **5.4 million snakebites, 1.8–2.7 million envenomings, and 81,410–137,880 deaths** annually worldwide [WHO fact sheet](#).

Intended Users

The WHO 2016 guideline is intended for medically trained personnel responsible for assessment and treatment of snakebite victims, including doctors, nurses, paramedics, dispensers, community health workers, medical students, and nursing students [WHO guidance PDF](#).

Scope and Objectives

The WHO 2016 clinical guideline addresses:

- Prevention and community awareness.
- First aid and avoidance of harmful traditional practices.
- Rapid referral and transport to health facilities.
- Clinical assessment and diagnosis of envenoming.
- Laboratory assessment, including clotting tests where relevant.
- Antivenom indications, administration, and management of reactions.
- Supportive care for neurotoxicity, shock, coagulopathy, kidney injury, and local tissue injury.

- Follow-up, rehabilitation, and health-system strengthening.

WHO frames snakebite as a medical emergency, particularly in rural areas, and emphasizes that rational antivenom use can substantially reduce mortality and morbidity [WHO publication page](#).

1.2 Key Recommendations

GRADE note: The supplied WHO 2016 guideline evidence snippets do not confirm formal GRADE use. Therefore, the GRADE ratings below are an evidence-informed assessment for this dynamic report, based on study design, consistency, directness, precision, and balance of benefits and harms.

Recommendation	GRADE Evidence Quality	Strength
Treat suspected venomous snakebite as a medical emergency and arrange rapid transport to a facility capable of diagnosis, resuscitation, antivenom administration, and monitoring.	Moderate	Strong
Reassure the patient, immobilize the bitten limb, minimize movement, remove constricting objects, and avoid delays in transport.	Low	Strong
Avoid harmful first-aid practices, including incision, excision, suction, "black stones," electric shock, and traditional remedies that delay care.	Low	Strong
Assess airway, breathing, circulation, neurological status, bleeding, local swelling, urine output, and shock immediately on arrival.	Moderate	Strong
Use clinical syndrome, local epidemiology, patient history, examination, and laboratory tests to determine whether envenoming has occurred and how severe it is.	Moderate	Strong
In low-resource settings, use the 20-minute whole blood clotting test to screen for venom-induced coagulopathy when laboratory coagulation testing is unavailable.	Moderate	Conditional
Administer appropriate antivenom early and in adequate dose when there are systemic signs of envenoming, clinically significant coagulopathy, neurotoxicity, shock, acute kidney injury, black urine, spontaneous bleeding, rapidly progressive swelling, necrosis risk, or dangerous bites such as digital bites by necrotizing species.	Moderate	Strong
Do not give antivenom for dry bites or trivial local symptoms without evidence of envenoming; observe and reassess.	Low	Strong
Administer antivenom intravenously by slow injection or infusion where possible, with adrenaline and resuscitation equipment immediately available.	Moderate	Strong
Be prepared to treat early antivenom reactions, including anaphylaxis, promptly with adrenaline and supportive care.	Moderate	Strong

Recommendation	GRADE Evidence Quality	Strength
Provide syndrome-specific supportive care: airway support and ventilation for paralysis, fluids/vasopressors for shock, blood products where indicated, renal monitoring and dialysis where needed, wound care, pain control, and rehabilitation.	Moderate	Strong
Avoid routine prophylactic antibiotics unless there is clinical evidence of secondary infection, necrosis, or another clear indication.	Low	Conditional
Strengthen prevention through community awareness, safe behavior, footwear/protective clothing, rapid referral pathways, trained health workers, reliable antivenom supply, and surveillance.	Low	Strong
Align antivenom procurement with WHO quality principles and local venomous snake epidemiology.	Moderate	Strong
Encourage national surveillance, burden assessment, affordable access to antivenom, rehabilitation, and integration with neglected tropical disease programmes.	Low	Strong

1.3 Guideline Development Methodology

Evidence Review Process

The WHO 2016 guideline is described as a revised and updated version of earlier WHO South-East Asia regional guidance. The guidance originated as a special issue in 1999, was later published as an independent regional document in 2011, and was revised with regional and global experts to incorporate new technical advances [WHO guidance PDF](#).

The supplied evidence did not verify that the 2016 WHO guideline used a formal GRADE process. It appears to rely on:

- Expert consensus.
- Regional clinical experience.
- Available observational and clinical evidence.
- Practical implementation considerations in resource-limited settings.

Consensus Method

The WHO 2016 guideline was revised with regional and global experts, with an emphasis on practical management by health workers at different levels of care [WHO guidance PDF](#).

Conflict of Interest Management

A snakebite-specific conflict-of-interest methodology document was not identified in the supplied evidence. General WHO governance documentation emphasizes that engagement with stakeholders should manage conflicts of interest to safeguard WHO’s integrity and independence [EB154/2024/REC/2](#).

II. Evidence Summary After WHO Latest Guideline Release (Evidence Cutoff Date: 29 April 2026)

2.1 New Evidence Overview

Search Strategy

Evidence was synthesized from WHO guidance, WHO technical pages, WHO news releases, global NTD reports, peer-reviewed reviews, diagnostic meta-analysis evidence, clinical trial registry data, national guidelines, and authoritative clinical sources from **16 August 2016** through **29 April 2026**.

Databases and Sources Searched

- WHO publications, WHO fact sheets, WHO neglected tropical disease pages, WHO news.
- WHO IRIS and World Health Assembly documentation.
- PubMed-indexed peer-reviewed articles.
- BMJ, Lancet-family sources, Frontiers in Immunology, PLOS Neglected Tropical Diseases.
- ClinicalTrials.gov.
- National and regional guideline sources.
- CDC/NIOSH occupational health guidance.
- Clinical toxicology and snakebite registry summaries.

Inclusion Criteria

Included evidence addressed at least one of:

- Snakebite diagnosis.
- First aid and referral.
- Antivenom efficacy, safety, access, or quality.
- Coagulopathy testing.
- Novel therapeutics.
- Snakebite surveillance, prevention, rehabilitation, or health-system response.
- WHO strategy or technical guidance.

Exclusion Criteria

Excluded evidence if:

- It was not related to snakebite envenoming.
- It lacked extractable clinical, diagnostic, therapeutic, or policy relevance.
- It was superseded by more current WHO guidance.
- It related to youth engagement but not snakebite, except as a methodology note.

2.2 Key New Studies

Study / source	Design	Key Findings	GRADE Quality
Lamb et al., 2021, 20-minute whole blood clotting test systematic review and meta-analysis	Diagnostic test accuracy systematic review and meta-analysis	Included 15 studies and 2,270 patients. For INR >1.4, 20WBCT sensitivity 0.84 and specificity 0.91; for fibrinogen <100 mg/dL, sensitivity 0.72 and specificity 0.94. Useful but imperfect bedside test in low-resource settings PubMed .	Moderate
BMJ "Managing snakebite," 2022	Clinical practice review	Supports baseline 20WBCT where laboratory testing is unavailable; emphasizes clinical assessment, monitoring, and timely antivenom when indicated BMJ .	Low to Moderate
Frontiers in Immunology, 2021, "Snakebite Envenoming Diagnosis and Diagnostics"	Narrative review	Diagnosis should focus on whether envenoming occurred, severity, and broad snake family when species-level diagnosis is not essential. Reviews PCR, ELISA, enzymatic assays, thermography, and other diagnostic research Frontiers in Immunology .	Low
FabAV copperhead randomized trial summarized in clinical review	Randomized controlled trial	FabAV improved limb function at 7, 10, and 14 days versus placebo; 75% recovered full limb use by day 31 versus day 57 in controls; benefit greater when given within 5.5 hours Medscape .	Moderate
Schaeffer et al. antivenom safety meta-analysis summarized in clinical sources	Meta-analysis	Acute adverse reactions after Crotalidae polyvalent immune Fab antivenom estimated at about 8%; serum sickness about 13% in the summarized meta-analysis Medscape .	Low to Moderate
North American Snakebite Registry antivenom safety data	Registry observational study	Acute adverse reactions after FabAV reported around 2.3% in adults and 2.7% in children; rash, hypotension, and bronchospasm each around 0.9% Medscape .	Low
Arizona FabAV-treated cohort summarized in clinical review	Observational cohort	Among 1,340 FabAV-treated patients, acute adverse reactions occurred in about 1.4%, with no reactions requiring antivenom discontinuation Medscape .	Low

Study / source	Design	Key Findings	GRADE Quality
Lancet Regional Health – Americas, 2025, late antivenom for coagulopathy	Observational effectiveness study	Examined late and very late antivenom administration for snakebite-induced coagulopathy in French Guiana; numerical outcomes not extractable from supplied summary Lancet Regional Health – Americas .	Low
ClinicalTrials.gov NCT04996264, BRAVO varespladib oral trial	Phase 2 multicenter randomized double-blind placebo-controlled trial	Evaluated oral varespladib for snakebite safety, tolerability, and efficacy; BRAVO Phase II reported completed in June 2023 ClinicalTrials.gov NCT04996264 , Impact Global Health .	Moderate for trial design; clinical certainty pending published outcomes
Landscape review of toxin-neutralizing therapeutics	Review	Identifies methyl-varespladib and unithiol as fast-moving repurposed candidates; suggests small molecules will likely need combination with antivenom, biologics, or other small molecules Landscape review .	Low
WHO TPPs for small molecule or engineered antibody therapeutics	WHO technical guidance	Defines minimum and optimum characteristics for novel snakebite therapeutics, including small molecules and engineered antibody biologics WHO publications page , WHO news .	Not applicable as clinical efficacy evidence; high relevance as technical guidance
South African Primary Healthcare guideline, 2024	National guideline	States most snakebite patients do not need antivenom; antivenom dosing same for adults and children; antibiotics seldom needed except secondary infection South African PHC guideline PDF .	Low to Moderate

2.3 Updated Clinical Evidence Synthesis

2.3.1 Burden and strategic priority

WHO estimates:

- **5.4 million** snakebites annually.
- **1.8–2.7 million** envenomings annually.
- **81,410–137,880** deaths annually.

- Approximately three times as many amputations and other permanent disabilities as deaths [WHO fact sheet](#).

WHO's global strategy aims to reduce snakebite mortality and disability by **50% by 2030**, through:

1. Empowering and engaging communities.
2. Ensuring safe and effective treatment.
3. Strengthening health systems.
4. Increasing partnerships, coordination, and resources [WHO fact sheet](#).

GRADE assessment:

The burden estimates are epidemiological and subject to underreporting, but consistent across WHO sources. Certainty for global burden magnitude is **Moderate**; recommendation to strengthen surveillance is **Strong**.

2.3.2 Prevention and community engagement

WHO states that raising community awareness about prevention is the most effective strategy for reducing snakebite morbidity and mortality [WHO guidance PDF](#).

Recommended prevention actions:

- Wear footwear and protective clothing in fields, forests, and at night.
- Use lights when walking in darkness.
- Avoid handling snakes, including apparently dead snakes.
- Improve sleeping arrangements in high-risk areas.
- Reduce rodent habitats near homes and food stores.
- Educate traditional healers and community leaders to promote rapid referral.

GRADE assessment: Low-quality direct evidence, but favorable benefit–harm balance and high feasibility. **Strong recommendation.**

2.3.3 First aid and referral

WHO recommends prompt transport to a health facility for suspected venomous snakebite and avoidance of traditional practices such as wound incision, excision, suction, and “black stones” [WHO NTD treatment page](#). CDC/NIOSH similarly advises emergency medical attention as soon as possible, keeping calm, neutral positioning, removing rings/watches, washing the bite, clean dressing, and marking swelling progression [CDC/NIOSH](#).

Actionable first aid:

1. Move away from the snake.
2. Reassure the patient and keep them still.
3. Immobilize the bitten limb.
4. Remove rings, watches, tight clothing.
5. Avoid alcohol, sedatives, cutting, suction, ice, electric shock, tourniquets, and herbal applications.
6. Transport urgently to the nearest equipped facility.
7. If safe, describe or photograph the snake, but do not attempt to catch or kill it.

GRADE assessment: Low-quality evidence, but strong indirect evidence of harm from delayed care and injurious practices. **Strong recommendation.**

2.3.4 Diagnosis and severity assessment

Clinical diagnosis remains central. The diagnostic priority is not always exact species identification; rather, clinicians should determine:

- Has envenoming occurred?
- What is the severity?
- Is the syndrome predominantly neurotoxic, hemotoxic/coagulopathic, cytotoxic/local necrotic, myotoxic, nephrotoxic, or mixed?
- What locally available antivenom covers likely species?

A 2021 Frontiers review states that where polyvalent antivenom is used, exact species or genus diagnosis may be less important than determining envenoming, severity, and broad snake family such as viper or elapid [Frontiers in Immunology](#).

Minimum assessment:

- Time and location of bite.
- First-aid already used.
- Symptoms: pain, swelling, ptosis, dysphagia, dyspnea, bleeding, dark urine, syncope.
- Vital signs.
- Local swelling progression.
- Neurological exam.
- Bleeding assessment.
- Urine output and color.
- 20WBCT or laboratory coagulation testing.
- Full blood count, creatinine, electrolytes, creatine kinase, urinalysis where available.

GRADE assessment: Moderate certainty for clinical syndrome-based diagnosis and monitoring; **Strong recommendation.**

2.3.5 20-minute whole blood clotting test

The 20WBCT has important post-2016 evidence. In Lamb et al.'s systematic review/meta-analysis:

- For INR >1.4:
 - Sensitivity **0.84**.
 - Specificity **0.91**.
 - SROC AUC **0.94**.
- For fibrinogen <100 mg/dL:
 - Sensitivity **0.72**.
 - Specificity **0.94**.
 - SROC AUC **0.93** [PubMed](#).

The test is most useful where formal coagulation assays are unavailable but is affected by operator training, glassware, ambient temperature, timing after bite, snake genus, and prior antivenom [PLOS PDF](#).

Updated recommendation:

- Use 20WBCT as a bedside screening tool for coagulopathy in low-resource settings.
- Repeat testing when initially negative but clinical risk persists.
- Use laboratory INR, fibrinogen, platelet count, and D-dimer where available.
- Do not rely on a single negative 20WBCT to exclude evolving coagulopathy.

GRADE assessment: Moderate diagnostic certainty; **Conditional recommendation** where laboratory testing is available, **Strong recommendation** where no laboratory alternative exists.

2.3.6 Antivenom: indications, timing, and administration

WHO states that antivenoms are the only specific treatment that can potentially prevent or reverse most effects of snakebite envenoming when given early and in adequate therapeutic dose [WHO NTD treatment page](#). Antivenoms are included in the WHO Model List of Essential Medicines [WHO activity page](#).

Indications for antivenom include:

- Shock.
- Spontaneous systemic bleeding.
- Uncoagulable blood or clinically significant coagulopathy.
- Neurotoxicity.
- Respiratory weakness or paralysis.
- Black urine / myoglobinuria / hemoglobinuria.
- Acute kidney injury.
- Rapidly progressive local swelling.
- Local necrosis risk.
- Bites by species known to cause severe local necrosis.
- Digital bites by necrotizing species.
- Severe systemic symptoms.

These indications are consistent with WHO-linked clinical summaries citing WHO guidance [Managing snakebite — PMC](#).

Administration principles:

- Use antivenom appropriate to local venomous species.
- Give intravenously by slow injection or infusion where possible.
- Do not delay indicated antivenom for skin testing.
- Have adrenaline and resuscitation equipment ready.
- Monitor for early reactions.
- Repeat antivenom based on persistent or recurrent systemic envenoming, coagulopathy, or neurotoxicity, according to local protocols and antivenom characteristics.

GRADE assessment: Moderate certainty overall; historical mortality benefit, strong biological plausibility, and RCT evidence in selected settings. **Strong recommendation.**

2.3.7 Antivenom safety

Post-2016 safety data show variable reaction rates:

- Meta-analysis summarized in clinical sources: acute adverse reactions after FabAV about **8%**, serum sickness about **13%** [Medscape](#).
- North American Snakebite Registry: acute adverse reactions about **2.3%** in adults and **2.7%** in children [Medscape](#).
- Arizona cohort of 1,340 FabAV-treated patients: acute adverse reactions about **1.4%**, none requiring discontinuation [Medscape](#).

Clinical implication: Antivenom reactions are real but usually manageable; they should not prevent antivenom use when clearly indicated.

GRADE assessment: Low to Moderate certainty for precise reaction rates due to heterogeneity; **Strong recommendation** to prepare for and treat reactions.

2.3.8 Novel therapeutics

The most important post-2016 development is WHO’s formal movement toward novel therapeutics.

WHO published **Target product profiles for small molecule or engineered antibody therapeutics for treatment of snakebite envenoming**, dated **31 December 2025**, and announced new guidance on **27 February 2026** [WHO publications page](#), [WHO news](#). These TPPs address small-molecule drugs and engineered antibody biologics and define minimum and optimum product characteristics for researchers, regulators, and manufacturers [InfoNTD](#).

Varespladib / varespladib-methyl:

- ClinicalTrials.gov lists NCT04996264, “Broad-spectrum Rapid Antidote: Varespladib Oral for Snakebite,” a multicenter randomized double-blind placebo-controlled Phase 2 trial [ClinicalTrials.gov NCT04996264](#).
- BRAVO Phase II trial completion was reported in June 2023 [Impact Global Health](#).

Current clinical recommendation: Novel therapeutics should not replace indicated antivenom outside approved indications or clinical trials as of 29 April 2026.

GRADE assessment: Moderate for trial design existence; low or very low for routine clinical effectiveness until peer-reviewed outcome data and regulatory approvals are available. **Conditional recommendation for use only in trials or approved protocols.**

2.4 Evidence Gaps and Future Research Needs

Evidence gap	Why it matters	Priority
Lack of globally standardized antivenom clinical trials	Many antivenoms are region-specific, and efficacy/safety evidence is uneven.	High

Evidence gap	Why it matters	Priority
Limited head-to-head antivenom comparisons	Needed for procurement and national formularies.	High
Limited evidence for late antivenom administration	Many patients present late; 2025 evidence is emerging but not definitive Lancet Regional Health – Americas .	High
Inadequate rapid diagnostic tests	Species/syndrome identification could improve antivenom selection and triage.	High
Variable 20WBCT performance	Training, equipment, and standardization are needed.	Medium
Pediatric dosing and outcomes	Children are high-risk, but many studies are adult-dominant.	High
Pregnancy-specific evidence	Snakebite in pregnancy requires more focused data.	Medium
Rehabilitation evidence	Disability prevention and long-term functional recovery are under-studied.	High
Novel therapeutics outcome data	Small molecules and engineered antibodies need robust clinical trial results.	High
Surveillance and burden data	Underreporting impairs antivenom planning and health-system response.	High
Youth and community engagement in snakebite guideline development	WHO has youth-engagement methodology in abortion-care guideline work, but no snakebite-specific case study was identified WHO webinar page .	Medium

2.5 Updated Recommendations Based on New Evidence

Original Recommendation	New Evidence Impact	Updated Recommendation
Prompt referral to hospital after suspected venomous bite	WHO, CDC/NIOSH, and clinical reviews reinforce urgency and harm from delayed care WHO NTD treatment page , CDC/NIOSH.	Maintain. Transport urgently to an equipped facility; avoid delays for traditional treatment or snake capture.
Avoid harmful first aid	Continued consensus that incision, suction, black stones, and similar practices are harmful or delay care.	Strengthen. Explicitly prohibit cutting, suction, ice, electric shock, tight tourniquets, black stones, and traditional remedies that delay referral.
Use clinical assessment and laboratory testing	Diagnostic reviews emphasize syndrome/severity assessment over exact species identification in many polyvalent-antivenom settings Frontiers in Immunology .	Maintain. Prioritize envenoming status, severity, syndrome, and local species epidemiology.

Original Recommendation	New Evidence Impact	Updated Recommendation
Use 20WBCT where appropriate	2021 meta-analysis gives moderate-quality diagnostic accuracy evidence PubMed .	Update. Use 20WBCT in low-resource settings, repeat if initially negative, and confirm with laboratory coagulation tests where available.
Give antivenom when indicated	WHO continues to identify antivenom as the only specific treatment; RCT and observational evidence support benefit in selected envenomations WHO NTD treatment page .	Maintain and strengthen. Give appropriate antivenom early for systemic envenoming or severe/progressive local envenoming; do not withhold solely due to fear of treatable reactions.
Do not give antivenom to all snakebite patients	National guideline evidence reinforces that many bites are dry or mild and do not require antivenom South African PHC guideline PDF .	Maintain. Observe stable patients without envenoming; reassess clinically and with repeat tests.
Prepare for antivenom reactions	Safety evidence quantifies reaction rates and supports readiness rather than avoidance Medscape .	Maintain. Administer only where adrenaline, airway support, and trained personnel are available whenever possible.
Use antibiotics selectively	Updated national guidance supports avoiding routine antibiotics unless secondary infection is present South African PHC guideline PDF .	Maintain. Do not use prophylactic antibiotics routinely.
Improve antivenom quality and access	WHO TPPs and WHA actions emphasize availability, accessibility, affordability, and regulatory standards WHA71 records .	Strengthen. National programmes should procure quality-assured, geographically appropriate antivenoms and maintain cold-chain/storage capacity.
Explore novel therapeutics	WHO 2025/2026 TPPs and Phase 2 varespladib trial show active pipeline development WHO publications page , ClinicalTrials.gov NCT04996264 .	New. Novel small molecules or engineered antibodies should be used only in approved indications, clinical trials, or formal protocols until efficacy and safety are established.
Strengthen surveillance and national programmes	WHA71 and WHO NTD reports emphasize surveillance, prevention, treatment, rehabilitation, affordability, and notifiable disease systems WHA71 records , Global report on neglected tropical diseases 2025 .	Strengthen. Countries should make snakebite reportable where feasible, track antivenom use/outcomes, and integrate rehabilitation.

Practical WHO-Based Care Pathway

A. Community / prehospital level

1. Move away from danger.
2. Reassure and immobilize.
3. Remove rings/tight items.
4. Avoid harmful first aid.
5. Transport urgently.
6. Record time of bite and symptom progression.

B. Primary facility level

1. Rapid ABC assessment.
2. Identify systemic envenoming signs.
3. Perform 20WBCT or coagulation tests.
4. Establish IV access.
5. Treat shock, pain, hypoxia, and airway compromise.
6. Give antivenom if indicated and available.
7. Refer if respiratory support, dialysis, surgery, or monitoring is unavailable.

C. Hospital level

1. Continue monitoring vitals, neurotoxicity, bleeding, swelling, urine output.
2. Repeat coagulation testing / 20WBCT.
3. Administer or repeat antivenom according to response.
4. Treat anaphylaxis immediately if it occurs.
5. Provide ventilatory support for paralysis.
6. Manage AKI, rhabdomyolysis, compartment concerns, necrosis, and infection.
7. Begin rehabilitation and follow-up.

Final Dynamic Guideline Position as of 29 April 2026

1. **Antivenom remains the cornerstone and only established specific treatment** for clinically significant snakebite envenoming when given early, in adequate dose, and matched to likely species [WHO NTD treatment page](#).
2. **Not every snakebite requires antivenom**; patients without evidence of envenoming require observation and reassessment.
3. **First aid should be simple and non-harmful**: immobilization, reassurance, and urgent transport; avoid incision, suction, black stones, and traditional practices that delay care.
4. **Diagnosis should be syndrome- and severity-based**, supported by local epidemiology and laboratory testing where available.
5. **20WBCT is useful in low-resource settings** but imperfect; repeat and confirm where possible.
6. **Antivenom reactions are manageable** and should not prevent use when antivenom is clearly indicated.
7. **Novel therapeutics are promising but not yet replacements for antivenom** outside trials or approved protocols.
8. **Health-system measures—surveillance, quality-assured antivenom access, trained staff, referral networks, rehabilitation, and community engagement—are essential** to reach WHO's 2030 target

of reducing snakebite mortality and disability by 50% [WHO fact sheet](#).